Module-X

Lecture-I

ATPG for Synchronous Sequential Circuits

1. Introduction

Till now we have been looking into VLSI testing, only from the context of combinational circuits. In this module we will study ATPG for sequential circuits. In the first lecture we will see the variations required in fault model, algebra and ATPG procedure in sequential circuits compared to combinational ones. Also we will compare the ATPG complexity in both the cases and show that testing sequential circuits is many folds more involved than testing combinational circuits. Following that in the next lecture we will introduce a special scheme called “scan chain” which ``modifies a sequential circuit into a virtual combinational one”. So, test patterns for a sequential circuit with scan chain can be generated with slight variation in ATPG algorithms for combinational circuits (D-algorithm, for example).

In this lecture we will first discuss the difference between sequential and combinational circuits from the context of ATPG and single stuck-at fault model. In this course we will concentrate only on sequential synchronous circuits with a single clock.

2. ATPG and testing: Sequential versus combinational circuits

Let us first see the basic architecture of a sequential circuit in Figure 1. In the figure it may be noted that there are three blocks, namely NSF, OFB and state flip-flop. The first two blocks are combinational circuits and the state flip-flop has sequential elements. So ATPG procedure for the NSF and OFB blocks should be similar to the ones already introduced in the previous lectures. However, it may be noted that compared to a standard combinational circuit, in case of NSF some inputs (state feedback) are not controllable and its outputs are not observable. Similarly, in case of the OFB, some of its inputs are non-controllable. When a circuit powers up, the flip-flops can have any value (0 or 1). So for ATPG of the combinational blocks in sequential circuits, we need to control (indirectly) the values in the nets which are outputs of flip-flops and observe (indirectly) the nets which are inputs to the flip-flops. Once indirect controllability and observability are achieved, ATPG for these combinational blocks (in sequential circuits) can be done using D-algorithm (or any other combinational APTG algorithm).

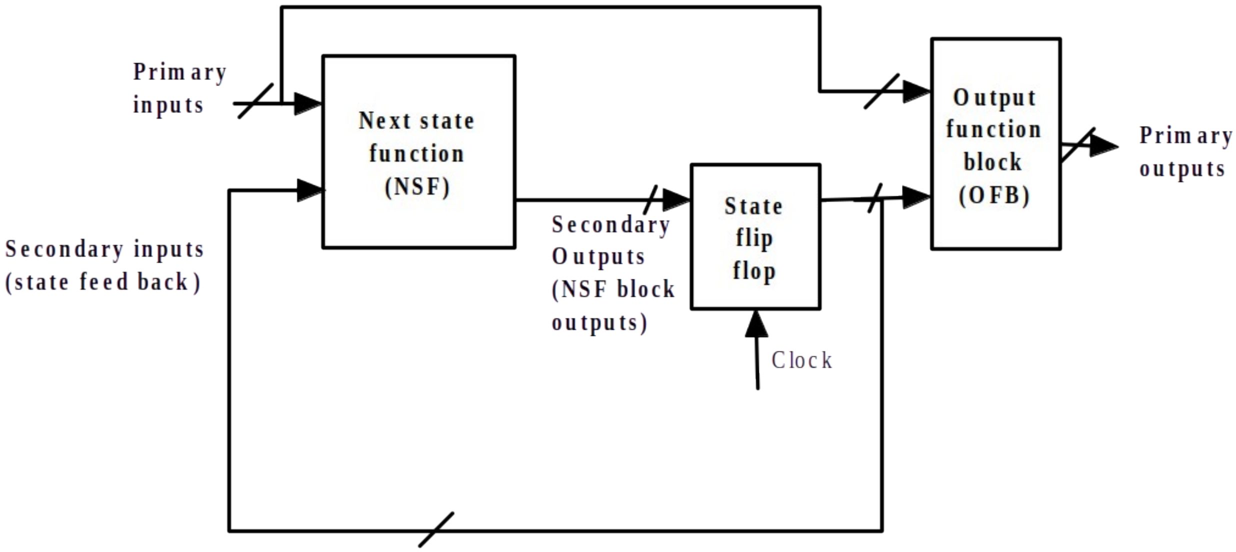


Figure 1. Basic architecture of a sequential circuit [1]

Before we illustrate how indirect controllability and observability can be achieved in these combinational blocks, we will discuss some assumptions in ATPG for single clock sequential circuits.

In single clock sequential circuits, each flip-flop is treated as 1-bit memory element with ONE common clock. For each primary input pattern and secondary input pattern (present state), the resulting primary output and secondary output (next state) patterns are produced. After a clock edge, the secondary output pattern (next state) is transferred to the output of the flip-flops (present state), which become new secondary inputs. Also, the primary outputs are updated. This activity occurs at each clock edge, and so it is called “synchronous” operation.

Single stuck at faults are assumed in the two combinational blocks namely, NSF and OFB. Internal faults of flip-flops are not modeled; their output and input faults are modeled as faults on input and output signals of the combinational blocks. In other words, flip-flops are treated as ideal memory elements and no faults are considered in the clock signal. Most of the time, D-flip-flops are used in VLSI designs. So in this course whenever we refer to a flip-flop we essentially mean a D-flip-flop.

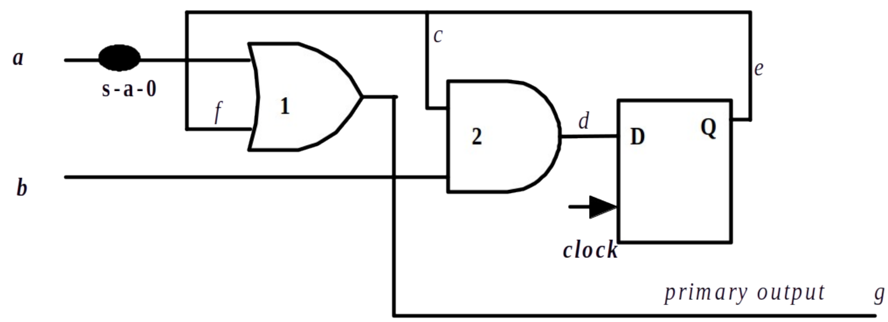


Figure 2. A simple sequential circuit with a stuck-at-0 fault

Let us consider a simple sequential circuit in Figure 2, with a s-a-0 fault at net *a*. If we compare this circuit with the block diagram shown in Figure 1, it may be noted that gate-1 corresponds to the OFB, gate-2 is for the NSF and D-flip flop is the memory element.

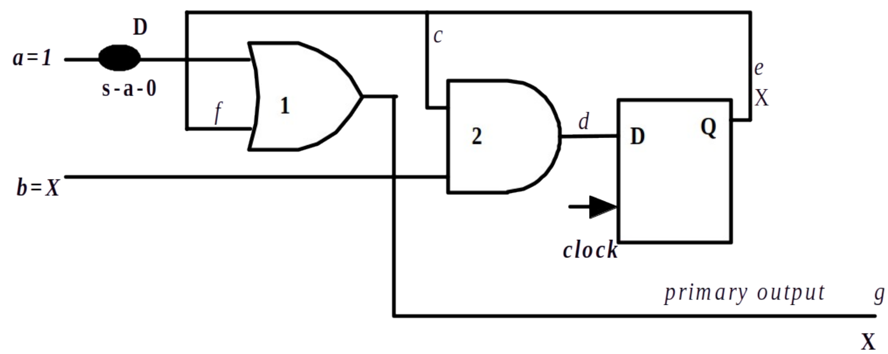


Figure 3. Problems in ATPG for the stuck-at-0 fault

To test the s-a-0 fault, simply, *a=1* and *f*=0, which results in D being propagated to the primary output. It may be noted that net *f* (secondary input) is driven by the output of the state flip-flop. As net *f* is not directly controllable, we cannot make it 0 and test the fault. As discussed, after power up, the flip-flop can have any signal value (marked as X). So testing the s-a-0 fault is not as simple as in case of combinational circuits. This problem is shown in Figure 3.

For ATPG of the fault, we need to indirectly control the net *f* to 0. Following that, we can apply the test pattern as *a*=1 and *b*=X, which propagates D to the primary output. From the circuit it may be noted that to make *f* to 0, we first need to make net *d=0,* which is the input to the flip-flop; following that a clock edge can transfer the value of *d* to *f* (in the next state). The process of making *f*=0 in a indirect manner (by state change) is shown in Figure 4.

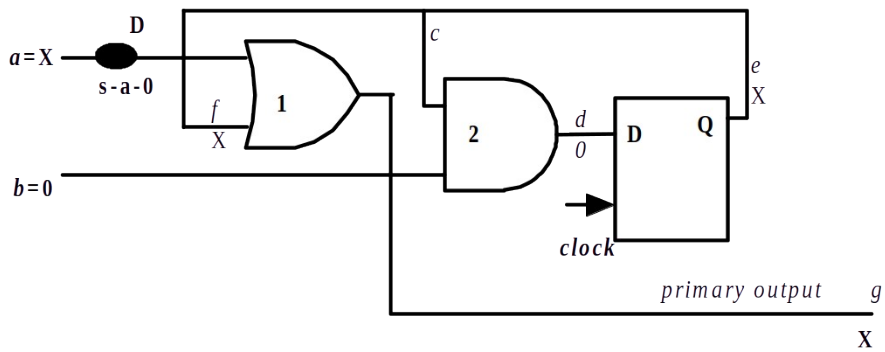


Figure 4. Indirect controlling of f to 0

After net f is made 0, then a=1 and b=X would propagate the fault effect (D) to the primary output; this is shown in Figure 5.

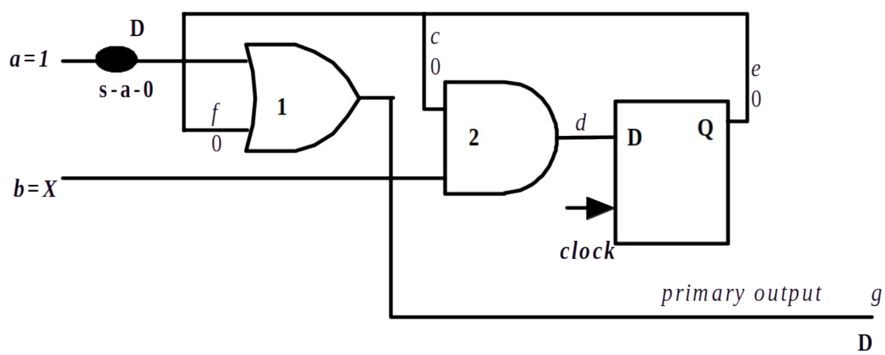


Figure 5. Test pattern for the s-a-0 fault

So, it may be noted that ATPG for combinational blocks in sequential circuits require more than one pattern. In this example, the first pattern is a=X, b=0 and clock edge followed by a=1 and b=X. In such cases, final pattern is according to combinational ATPG (that sensitize the fault and propagates the effect to a primary output) and all other initial patterns are to bring the secondary inputs to their required value.

ATPG for a fault in the NSF block is more complex than the ones in the OFB. In case of a fault in the NSF we need to indirectly control the secondary inputs and in addition, propagate the fault effect to the primary output via OFB. This is illustrated using a s-a-1 fault at net c in the circuit (Figure 6). First, primary input b=0 (a=X) and a clock edge is applied; this makes *e*=0 (and also *c*=0) after the edge. It may be noted that even in presence of the s-a-1 fault, *b*=0 and a clock edge makes *c*=0; this sensitizes the fault ($\overline{D}$ at *c*). In the next step, *b*=1 (*a*=X) and a clock edge would propagate the fault effect ($\overline{D}$) to the output of the flip-flop (i.e, secondary input of the OFB). Finally, *a*=0 (*b*=X) would propagate fault effect to the primary output. Above mentioned three steps are shown in Figure 6.

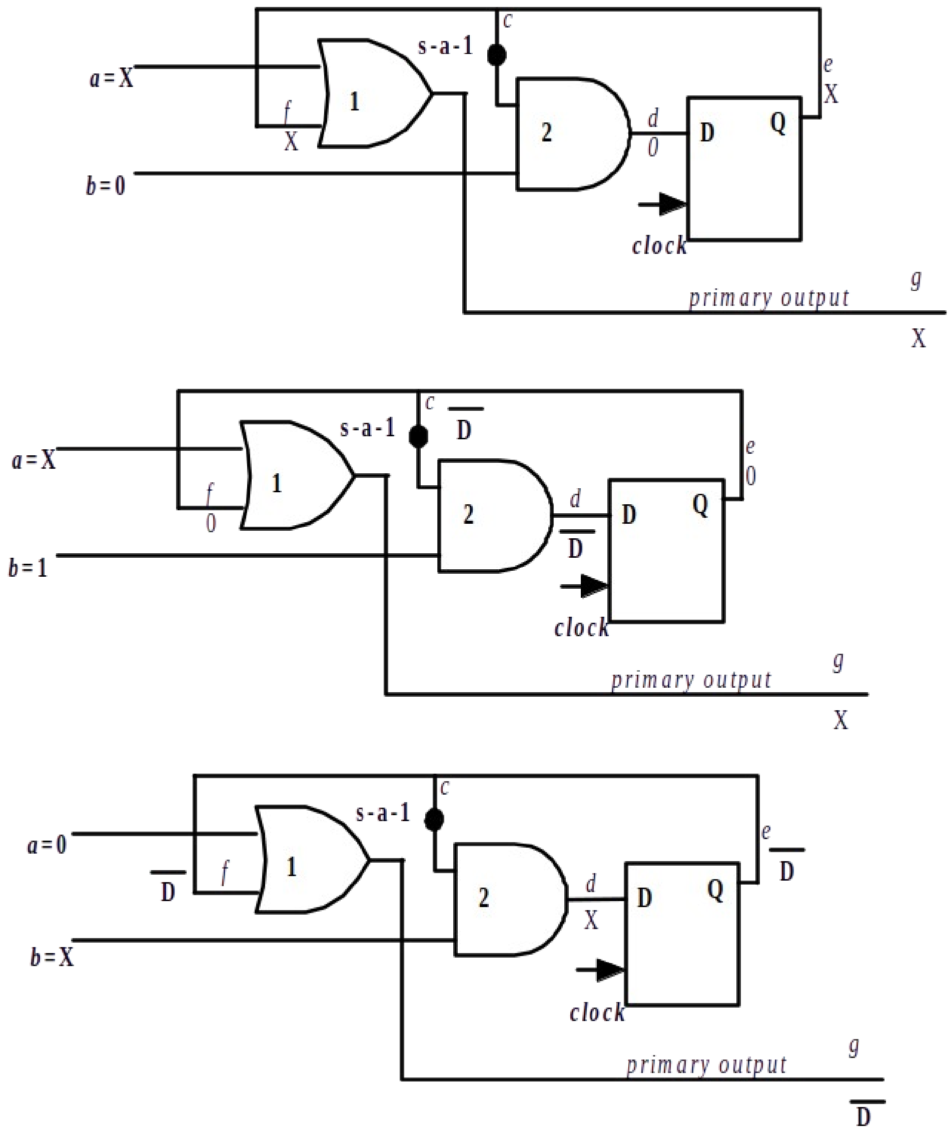


Figure 6. ATPG for s-a-1 fault in the NSF block

2.1 Algebra for ATPG of sequential circuits

In the last module we saw that Boolean algebra is not suitable for ATPG of combinational circuits; Routh's (5 value) algebra was found appropriate for ATPG algorithms. Now we will find whether this 5 value algebra suffices for ATPG of sequential circuits or we need to move to even higher orders. Let us try ATPG for a s-a-1 fault in the sequential circuit shown in Figure 7 using 5 value algebra.

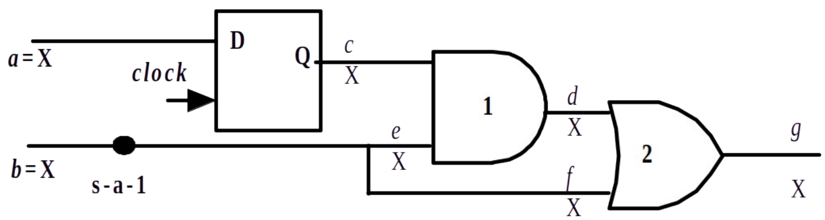


Figure 7. A simple sequential circuit with a stuck-at-1 fault

Fault effect can be propagated by two paths "*e-d-g*" or "*f-g*". Let us take the shorter path (*f-g*). For this case we need net *d*=0. Also, to sensitize the fault we need to apply 0 at *b*. As net *d* is not directly controllable, we need to make *a*=0 (*b*=X) and apply a clock pulse. Next, if we apply *b*=0 (a=X), the fault effect $\overline{D}$ would appear in the primary output. This process is shown in Figure 8.

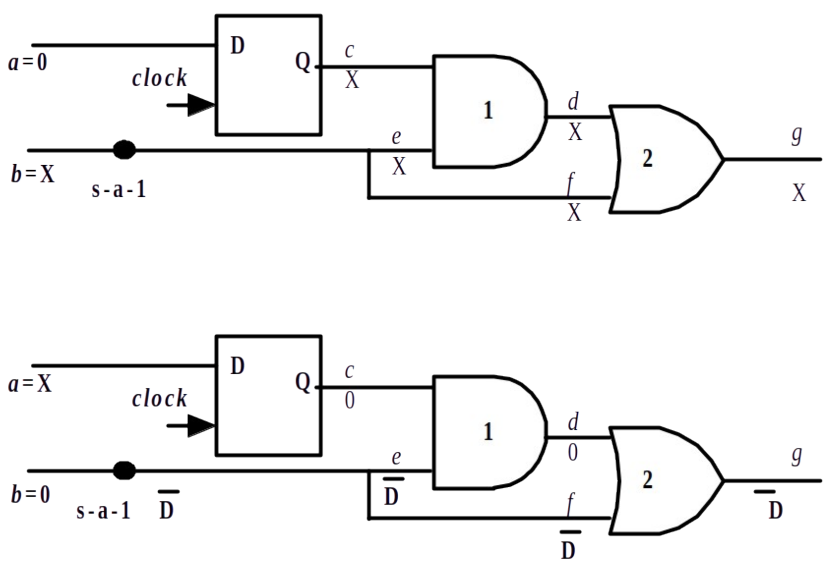


Figure 8. ATPG for s-a-1 fault (using Routh's algebra)

Now, we will see an interesting thing. If we use higher order algebra, then only one pattern can test the fault. Let us see the case if *b*=0 (*a*=X). Fault is sensitized as $\overline{D}$ and *e*=*f*=$\overline{D}$. Now as the secondary input (*c*) is not controlled, *c*=X. By 5 value algebra, *d*=X and so fault effect cannot be propagated to the output. However, if we observe carefully at net *d*, we will note that if the fault is there then *d*=X (as value is determined by the flip-flop at start up of circuit). However, if fault is not there then *d*=0 (as *e*=0). So we can mark *d* as 0/X. Now, *f*=$\overline{D}$, which in turn makes g=$\overline{D}$; $\overline{D}$ (=0/1) OR 0/X = 0/1 (=$\overline{D}$). So fault can be detected using a single pattern *b*=0 (*a*=X). All these steps are shown in Figure 9.

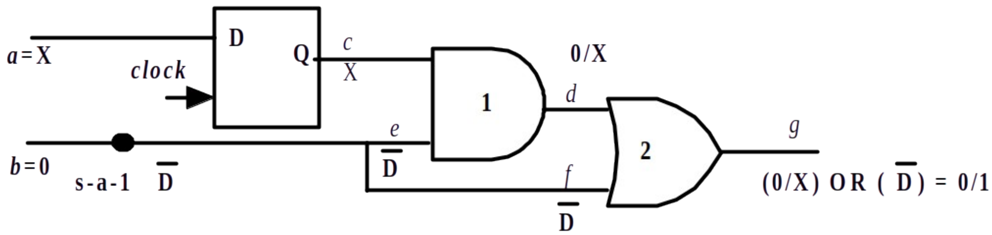


Figure 9. ATPG for s-a-1 fault using higher order algebra

The following points are to be noted

- In the example, higher order algebra is used, as a net is marked as 0/X, which is not available in 5 value algebra.
- Higher order algebra improves efficiency of ATPG of sequential circuits. As higher order algebra reduces the number of input (primary and secondary) lines to be controlled, there is reduction in the number of steps (in terms of clock edges and test patters) to control the secondary inputs (or make the NSF block outputs observable via OFB). However, it does not guarantee that ATPG will not require controlling the secondary inputs and propagating the fault effect to the primary output via OFB. This example was a special case where controlling the secondary inputs were not required.
- Higher order algebra will also reduce the number of lines to be controlled in ATPG of a combinational circuit. However, it is not applied as computational complexity rises with increase in order of the algebra and inputs are easily controllable in combinational circuits.

Table 1 illustrates a 9 value algebra which is used for ATPG of combinational blocks in sequential circuits.

Table 1. Nine value algebra

Symbol	**Implication**	**Normal Circuit**	**Faulty Circuit**
0	(0/0)	0	0
1	(1/1)	1	1
X	(X/X)	X	X
D	(1/0)	1	0
$\overline{D}$	(0/1)	0	1
G0	(0/X)	0	X
G1	(1/X)	1	X
F0	(X/0)	X	0
F1	(X/1)	X	1

3. ATPG of sequential circuits: Time frame expansion method

ATPG for (combinational blocks in) sequential circuits involves the following two broad steps:

1. Replace all the flip-flops by nets. Perform D-algorithm on the virtual combinational circuit and find the values of primary and secondary inputs.
2. Using multiple steps (primary inputs, secondary inputs and clock pulses) to apply the required signals to the secondary inputs (determined in Step 1).

The circuit shown in Figure 10 will be used to illustrate ATPG for sequential circuits using the time frame expansion approach.

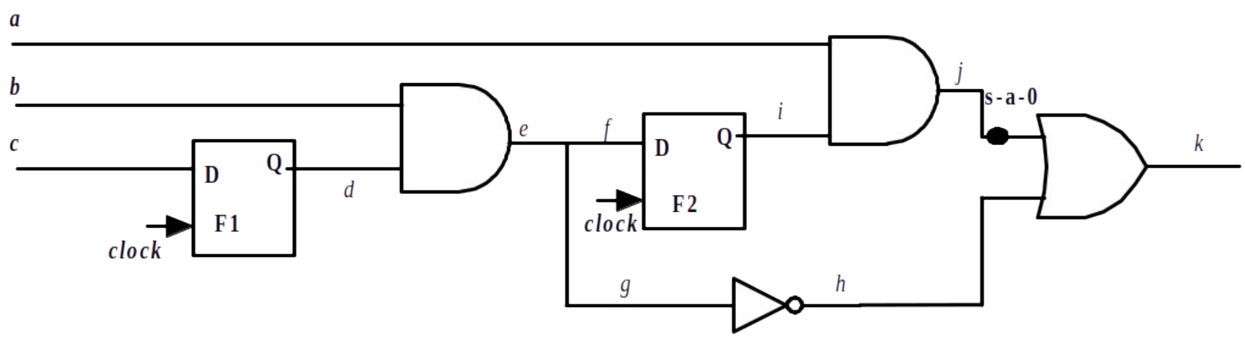


Figure 10. Sequential circuits with a s-a-0 fault

In the circuit we may note that nets *a*,*b*,*c* are primary inputs and nets *d*,*i* are secondary inputs. So if we replace the flip-flops with nets we get the circuit shown in Figure 11; such replaced nets are marked by doubled sided arrows. Also in the same figure we determine the test pattern to detect the s-a-0 fault as: *a*=1, *b*=1, *c*=1, *d*=1,*i*=1 and output is **D.** *a*=1, *b*=1, *c*=1 can be obtained by directly controlling the primary inputs. However, *d*=1,*i*=1 corresponds to secondary inputs which need to be achieved by setting primary inputs appropriately and clock edges, in steps previous to applying the "final" primary input as *a*=1, *b*=1, *c*=1.

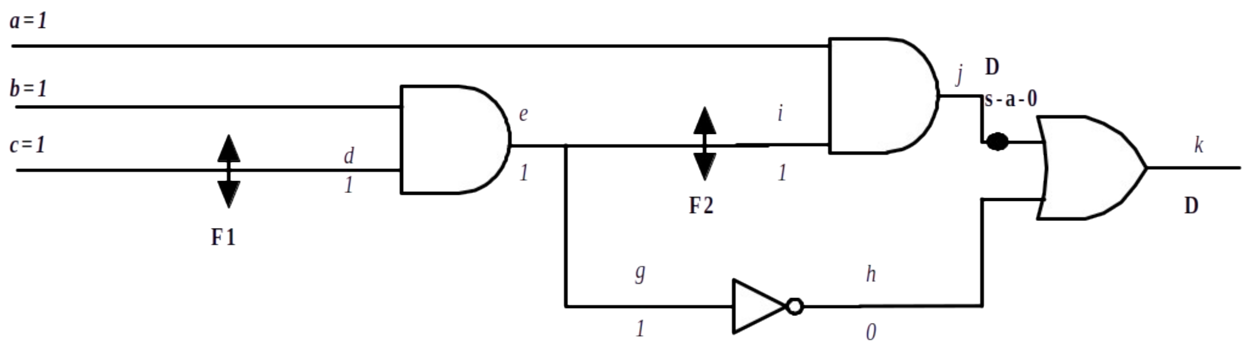


Figure 11. Sequential circuit of Figure 10 with flip-flops replaced by nets

Before we study in detail the ATPG for the circuit given in Figure 10, we disuses the following definitions and conditions.

Definition 1: Sequential depth of a flip-flop:

If the output of a flip-flop can be controlled by only primary inputs (and a clock pulse) it has sequential depth of 1. In the circuit of Figure 10, flip-flop F1 has sequential depth of 1 as it is controllable by primary input(s) *c*. A flip-flop has a sequential dept of d_{seq} if its output is dependent on primary inputs and at least one flip-flop of depth d_{seq}*-1*. In the example, flip-flop F2 is dependent on primary input *b* and output of flip-flop F1 (via net *e*); so sequential depth of F2 is 2.

Definiation2: Non-cyclic Circuit

A sequential circuit is non-cyclic if there is no flip-flop whose input is dependent on its output. For example, in the circuit of Figure 10, input of F1 is dependent on only primary inputs. On the other hand, input of F2 is dependent on primary inputs (net *b*) as well as the output of F1 (net *d*). So the circuit non-cyclic. An example of a cyclic sequential circuit is given in Figure 12. In this case input of F1 is dependent on its own output (and primary input). It may be noted that in cyclic circuits we may not be able to control the secondary inputs; this concept will be elaborated in an exercise problem. A flip-flop whose input is dependent on its own output is called a cyclic flip-flop.

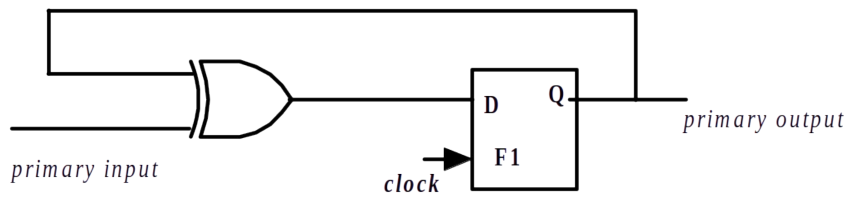


Figure 12. Cyclic sequential circuit

Property 1. The secondary inputs of a cycle free sequential circuit of depth d_{seq} can be brought to controllable value is at most d_{seq} primary input patterns and clock pulses.

Proof Idea: The proof is obvious. All flip-flops with $d_{seq}=1$ can be controlled by setting primary inputs and a clock pulse. Now, as all flip-flops with $d_{seq}=1$ have been set, we can control flip-flops with $d_{seq}=2$ by setting primary inputs and a clock pulse. In this order, if there are flip-flops with $d_{seq}=n$, we require at most n primary input patterns and n clock pulses. The proof idea for a circuit with $d_{seq}=2$ is shown in Figure 13.

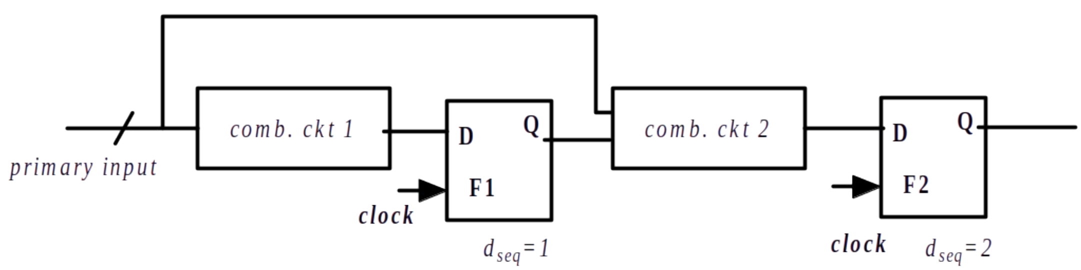


Figure 13. Illustration of the proof idea of Property 1

Now we will illustrate ATPG for the s-a-0 fault in the circuit of Figure 10, using time fame expansion approach and nine value algebra. For the circuit, d_{seq} =2; so we require 2 primary input patterns and clock pulses to set the secondary inputs (*d*,*i*). As determined from the virtual combinational circuit (Figure 11), test pattern for the fault is *a*=1, *b*=1, *c*=1, *d*=1,*i*=1. So we need set *d*=1,*i*=1 and finally apply *a*=1, *b*=1, *c*=1. This can be achieved in three time steps:

- **Time Step-1:** Set primary inputs such that F1 is set appropriately which (along with primary inputs) can enable F2 to be 1 in Time Step-2. As *i* is to be 1 in Time Step-2, so *d* is to be made 1 by making primary input *c*=1 (in Time Step-1). Other primary inputs are don't cares in this step. Finally a clock pulse is given.
- **Time Step-2:** Net *d*=1 from settings in Step-1. Now we need to appropriately set primary inputs such that F2 is 1 and along with it F1 is also 1, in Time Step-3. To make F2=1 (having set *d*=1) we need to have *b*=1. Also, to have F1=1 we need to have *c*=1. The other primary input *a* is don't care. Finally a clock pulse is given.
- **Time Step-3:** So we have *d*=1 and *i*=1, from the setting in Time Step-2; equivalently F1=1 and F2=1. Now as per the test pattern given in Figure 11, primary input *a*=1 and *b*=1 would result in **D** at the output**.**

So the test pattern for the fault is

(i) *a*=X,*b*=X,*c*=1 (clock pulse), (ii) *a*=X,*b*=1,*c*=1 (clock pulse) and (iii) *a*=1,*b*=1,*c*=X.

Now we will discuss in brief the procedure of drawing the circuit structure for sequential ATPG using time frame expansion approach for the running example.

The ATPG space is divided into d_{seq} + 1 sub spaces. The subspaces are numbered from - d_{seq} to 0. 0^{th} sub-space represents Time Step=d_{seq} + 1, where all secondary inputs are set. -d_{seq} sub-space represents the first time step when the circuit has just powered up and all secondary inputs are initialized to X. Following that, each sub-space represents change of signal values in the circuit with progress in time steps (i.e., primary input pattern and clock pulses). The basic block diagram of this procedure is shown in Figure 14. Figure 15 shows time steps for ATPG of the s-a-0 fault in the circuit of Figure 10.

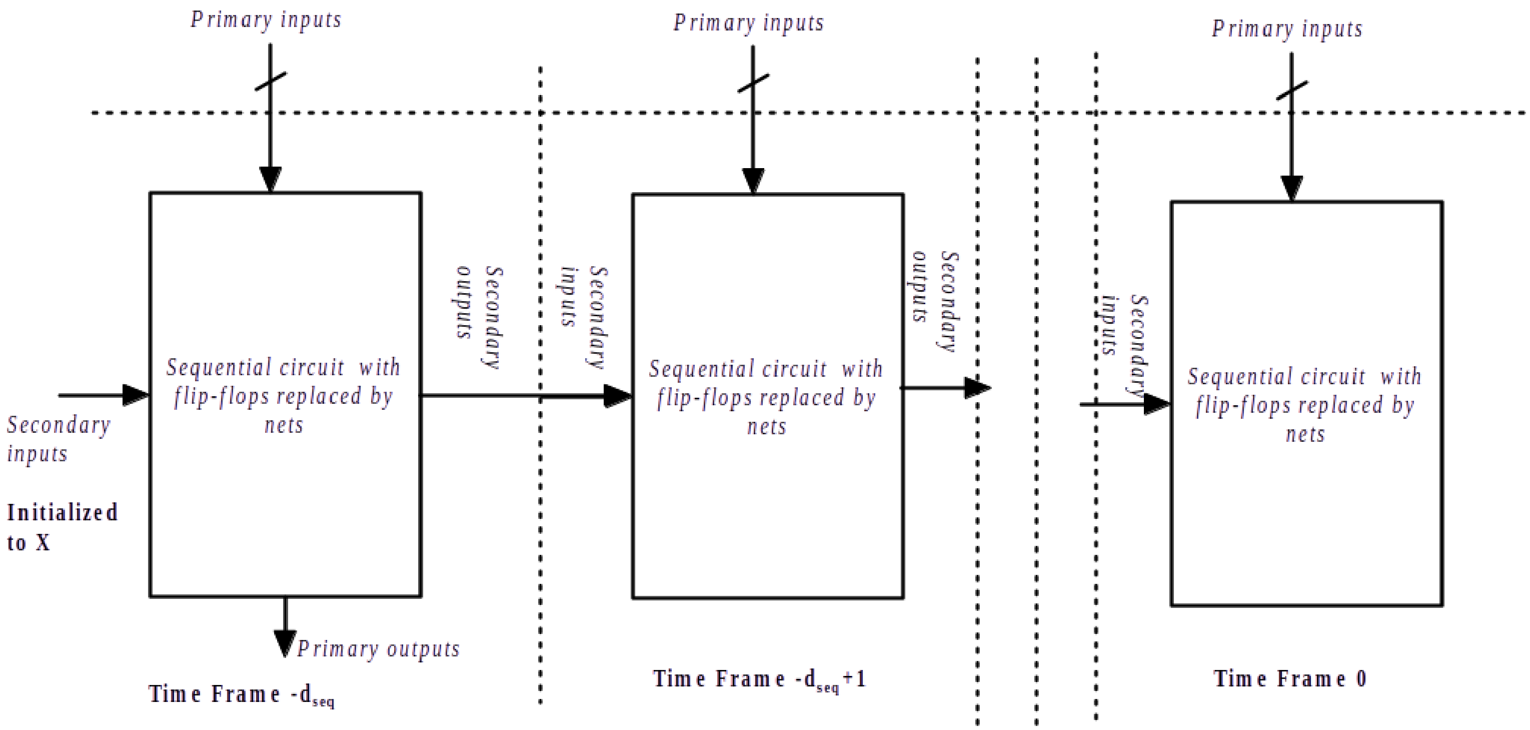


Figure 14. Time Steps for sequential ATPG using time frame expansion method

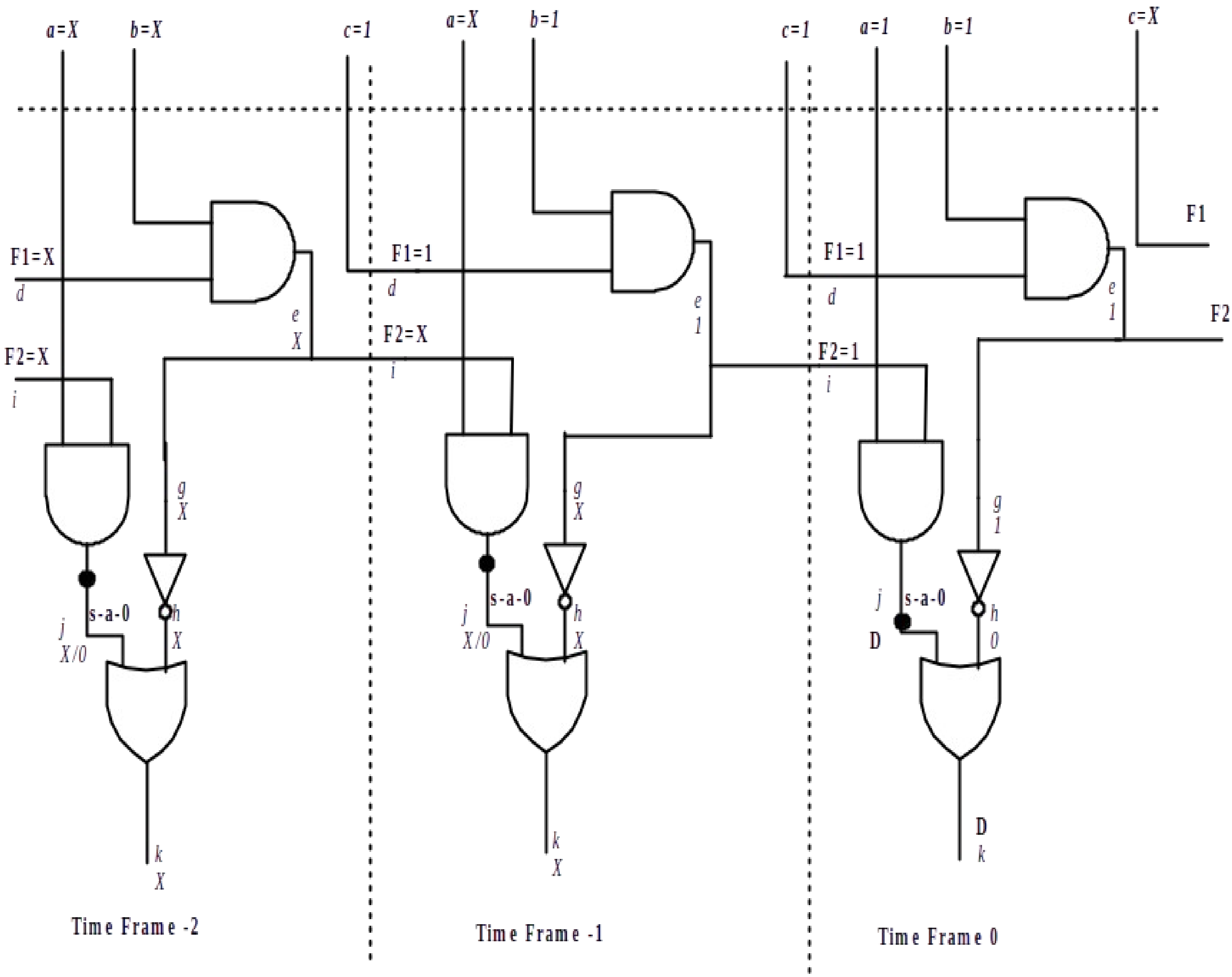


Figure 14. ATPG for the s-a-0 fault in the circuit of Figure 10: time frame expansion approach

3. Conclusions

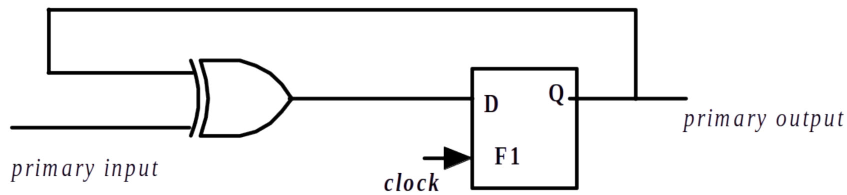
In this lecture we have seen that for ATPG and testing sequential circuits (combinational blocks) we need multiple patterns and clock pulses. So applying the patterns for testing a sequential circuit involves much more test time than a combinational circuit. Also the complexity of ATPG algorithms for sequential circuits is higher than of combinational ones. The problem was the controllability of secondary inputs and observability of secondary outputs. In the next two lectures we will study a DFT technique called "scan-chain" which eases sequential circuits testing and ATPG by making controllability and observability much simpler.

References

1. Mano, M. Morris,. Digital Design, 2/e. Prentice-Hall of India. 1995.

Questions and Answers

1. Can ATGP and testing be performed for any fault in the XOR gate of circuit given below



Answer:

The circuit is cyclic as input of F1 depends on its own output. When the circuit starts, output of F1 is X. Now, input of F1 is inversion of the primary input value, if F1 at startup was 1. Similarly, input of F1 is the primary input value if output of F1 at startup was 0. So it is obvious that as X is unknown we cannot set F1 according to need for ATPG. So ATGP and testing cannot be performed for any fault in the XOR gate of the circuit.